Fundamentals and K3D

Introducing Kubernetes

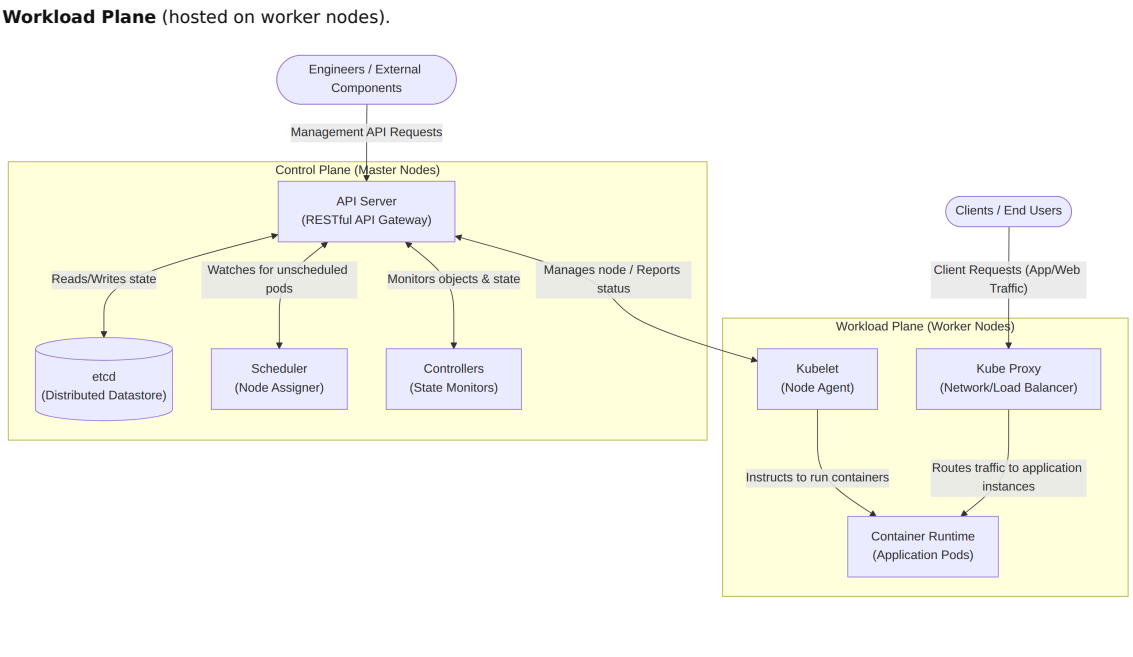
Kubernetes (Greek for *helmsman*) is a software system for automating the deployment and management of complex, large-scale application systems composed of computer processes running in containers. While you act as the captain deciding the system's overall direction, Kubernetes acts as the helmsman that steers the applications and reports on their operational status.

Core Benefits of Kubernetes

1. **Infrastructure Abstraction** – Kubernetes hides the details of the underlying hardware, networks, and computers from the users and applications.
2. **Declarative Deployment** – Users describe the desired state of an application via a single manifest, and Kubernetes automatically takes the necessary steps to turn that description into a running application.
3. **Automated Management** – The system assumes daily management tasks, such as automatically restarting failed applications or moving them to healthy nodes in the event of hardware failures. All this without **modifying the application** (e.g. by using Resience4J).
4. **Improved Hardware Utilization** – By dynamically deciding where to place each application based on available resources, Kubernetes tightly packs applications together to run more workloads on fewer servers.
5. **Portability** – Because applications interact with standardized Kubernetes APIs rather than proprietary cloud provider APIs, systems can be easily moved between local data centers and various cloud providers.

The Kubernetes Architecture

A Kubernetes cluster is fundamentally divided into two groups of machines: the **Control Plane** (hosted on master nodes) and the **Workload Plane** (hosted on worker nodes).


Engineers / External Components
Management API Requests
Control Plane (Master Nodes)
API Server (RESTful API Gateway)
Reads/Writes state
Watches for unscheduled pods
Monitors objects & state
Manages node / Reports status
etcd (Distributed Datastore)
Scheduler (Node Assigner)
Controllers (State Monitors)
Clients / End Users
Client Requests (App/Web Traffic)
Workload Plane (Worker Nodes)
Kubelet (Node Agent)
Kube Proxy (Network/Load Balancer)
Instructs to run containers
Routes traffic to application instances
Container Runtime (Application Pods)

Control Plane Components (Master Nodes)

The Control Plane serves as the brain of the system, holding and controlling the state of the cluster without running the actual applications.

Component	Function
API Server	Exposes the RESTful Kubernetes API used by engineers and components to create and manage objects.
etcd	A distributed datastore that persists all the objects created via the API Server.
Scheduler	Decides which specific worker node should run each newly created application instance.
Controllers	Monitor the API for new objects and perform the necessary operations to bring those objects to life.

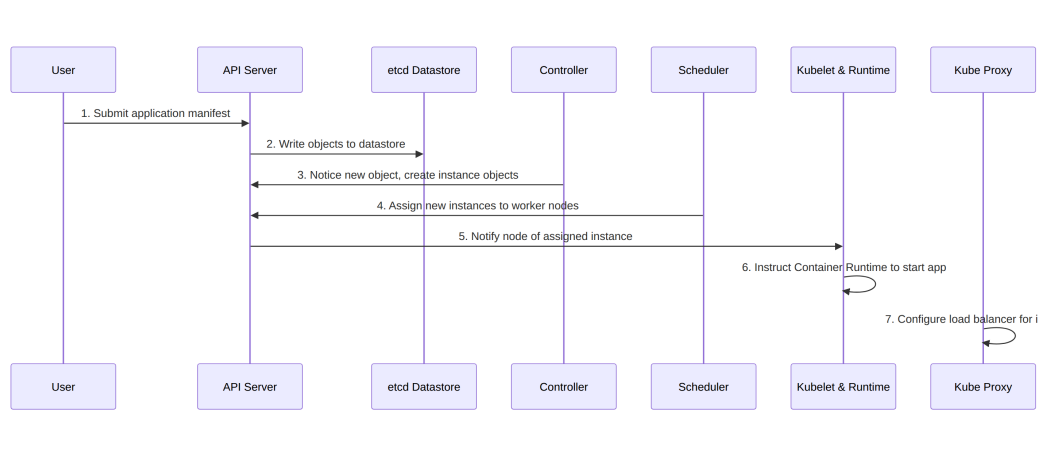
Workload Plane Components (Worker Nodes)

The Workload Plane consists of the worker nodes where your actual applications (workloads) are executed.

Component	Function
Kubelet	An agent that communicates with the API server, manages local applications, and reports node and application status.
Container Runtime	Software (usually `podman`) that physically runs the application containers as instructed by the Kubelet.
Kube Proxy	Sets up load balancers and routes network traffic between different application instances.

Application Deployment Flow

When you deploy an application, a sequence of automated events occurs across the cluster components.


User
API Server
etcd Datastore
Controller
Scheduler
Kubelet & Runtime
Kube Proxy
1. Submit application manifest
2. Write objects to datastore
3. Notice new object, create instance objects
4. Assign new instances to worker nodes
5. Notify node of assigned instance
6. Instruct Container Runtime to start app
7. Configure load balancer for instances

Deployment Environments

Organizations must decide where to physically host their Kubernetes clusters, each offering different operational characteristics.

1. **On-Premises** – Runs directly on your organization's bare-metal machines or virtual machines. This is often required by strict regulations but is harder to scale horizontally on short notice.
2. **Cloud** – Runs on a cloud provider's infrastructure. This offers high elasticity, allowing Kubernetes to provision or destroy virtual machines on demand to optimize resources and costs.
3. **Hybrid Cloud** – Runs locally but is configured to "spill over" into the cloud during temporary periods of peak load.

Challenges and Considerations

While powerful, Kubernetes is not the right choice for every organization or application architecture.

1. **High Complexity** – Managing a production-ready cluster requires intimate knowledge of its inner workings. Inexperienced teams are strongly advised to use managed Kubernetes-as-a-Service offerings (like GKE, EKS, or AKS) rather than deploying it themselves.
2. **Not for Monoliths** – If an application is a large, tightly-coupled monolith, Kubernetes provides no architectural benefits.
3. **Overkill for Small Systems** – Systems containing fewer than five microservices may find the added complexity and resource overhead of Kubernetes outweighs its automation benefits.
4. **Initial Cost Investment** – Introducing Kubernetes requires upfront investments in engineering time for training, building new tools, and procuring additional computing resources for the cluster overhead itself.

> ❓ ***When to use Kubernetes?***
> *Kubernetes excels in high complexity scenarios like a geographically distributed hybrid cloud.*
> *Consider using **simpler solutions** in less complex scenarios (like **Docker Swarm** for a **small on-premise private cloud** or **Slurm** for **batch HPC applications**).*

`kubectl`

`kubectl` (often pronounced `kube-control` or `kube-cuddle`) is the primary CLI tool for managing a Kubernetes cluster. You can install it by running the following command:

```
# GNU/Linux
curl -LO "https://dl.k8s.io/release/$(curl -L -s
https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"

# macOS
curl -LO "https://dl.k8s.io/release/$(curl -L -s
https://dl.k8s.io/release/stable.txt)/bin/darwin/arm64/kubectl"
```

`kubectl` features native autocompletion, which saves a lot of typing. To configure it, follow the official guides for Linux or macOS.

How `kubectl` works

`kubectl` is a user-friendly wrapper for the Kubernetes API server. Under the hood, it is simply a REST client that sends HTTP requests to the cluster's Control Plane. It communicates with the API server specified in the YAML configuration file located at `~/.kube/config`, or at the path specified by your `KUBECONFIG` environment variable. Usually, you don't have to write this file manually; tools like `k3d`, `minikube`, and cloud CLIs (like gcloud) will generate and modify it automatically.

How to install Kubernetes without having a cluster?

You don't need a full-fledged cloud environment or dedicated servers to learn Kubernetes. Several software tools allow you to run test clusters directly on your local machine:

Tool	Description	Pros	Cons
Docker Desktop	A built-in feature of the Docker Desktop GUI that spins up a single-node cluster.	Simple one-click setup via GUI; easy installation on Windows/macOS.	Very heavy on system resources; only supports single-node clusters.
kind *(Kubernetes in Docker)*	A CLI tool that runs standard Kubernetes by spinning up Docker containers to act as the "nodes".	Supports multi-node clusters; runs the full, standard version of Kubernetes.	Network traffic inspection between nodes (e.g., using Wireshark) can be difficult.
Minikube	A mature CLI tool that runs a cluster inside a separate VM or using containers.	Great out-of-the-box experience; highly supported with many built-in addons.	Can be slower and resource-heavy if using the VM driver.
k3d	A CLI wrapper that runs `k3s` (a highly efficient, lightweight Kubernetes distribution by Rancher) inside Docker.	Extremely lightweight (low CPU/RAM usage); fast startup; supports multi-node clusters.	Uses `k3s` instead of standard k8s, which is perfectly fine for 99% of use cases but strips out some legacy/enterprise features.

We will use `k3d` because of its features.

To install `k3d` run the following command:

```
# GNU/Linux
curl -s https://raw.githubusercontent.com/k3d-io/k3d/main/install.sh | bash

# macOS
brew install k3d
```

On `bash` and `zsh`, `k3d` features native autocompletion.
Follow this guide to configure it: k3d autocompletion.

Start a simple cluster

To start a cluster with `k3d` run the following command.

```
k3d cluster create CLUSTER_NAME -s 1 -a 3
```

Here's a breakdown of this command:

- `cluster create CLUSTER_NAME` : creates a cluster with specified name (default name is `k3s-default`)
- `-s N` : *server*, number of nodes in control plane
- `-a N` : *agent* number of worker nodes

To list the clusters on your device:

```
k3d cluster list
```

To stop/start an existing cluster:

```
k3d cluster stop
k3d cluster start
```

To remove a cluster and its files:

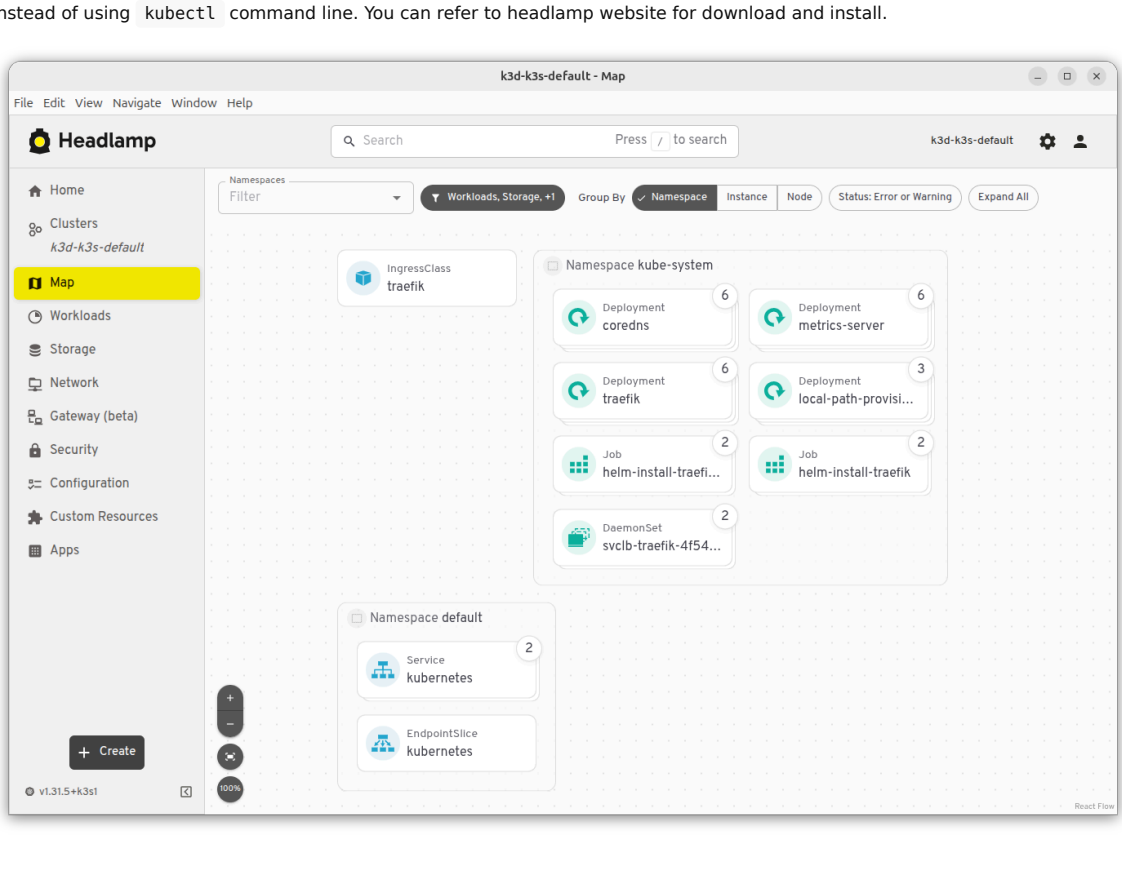
```
k3d cluster delete CLUSTER_NAME
```

You can verify that your cluster has successfully started by running

```
kubectl cluster-info
```

Headlamp

Sometimes, it can be useful to check the cluster status from a GUI.
Headlamp is a dashboard for Kubernetes which allows you to see the status of the cluster and run some routine commands easily instead of using `kubectl` command line. You can refer to headlamp website for download and install.


k3d-k3s-default - Map
File Edit View Navigate Window Help
Headlamp
Search
Press / to search
k3d-k3s-default
Home
Clusters
k3d-k3s-default
Map
Workloads
Storage
Network
Gateway (beta)
Security
Configuration
Custom Resources
Apps
Create
v1.31.5+k3s1
Namespaces
Filter
Workloads, Storage, +1
Group By
Namespace
Instance
Node
Status: Error or Warning
Expand All
IngressClass
traefik
Namespace kube-system
Deployment
coredns
6
Deployment
metrics-server
6
Deployment
traefik
6
Deployment
local-path-provisi...
3
Job
helm-install-traefi...
2
Job
helm-install-traefik
2
DaemonSet
svclb-traefik-4f54...
2
Namespace default
Service
kubernetes
2
EndpointSlice
kubernetes
100%
React Flow

Docker

Before we start make sure you have Docker installed, a Docker Hub account and that you have run `docker login`. Indeed, you will need to create containers and upload them on a registry to make them usable from Kubernetes' nodes.

Brief guide if you have never done this.

When building an image, make sure to tag it:

```
docker build . -t DOCKER_HUB_USERNAME/IMAGE_NAME:version
```

Once you have built an image, you can push it to the registry with

```
docker push DOCKER_HUB_USERNAME/IMAGE_NAME:version
```